

Youssef Elhagrasy

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EDUCATION

3rd Year Engineering Physics - University of British Columbia
Specialization in Electrical/Computer Engineering, Dean's Honour List

Vancouver, Canada
 Sep. 2022 – May 2026

EXPERIENCE

Product Engineer Co-op

January 2024 – April 2024

Microchip

Burnaby, Canada

- Created python test scripts to test TX jitter across a family of chips and their firmware.
- Tested and verified the internal interference of the chips using a Signal Source Analyzer (SSA).
- Experimented and reported TX jitter measurements using an 80 GHz oscilloscope.
- Measured VCO coupling based on architecture across a family of chips for mixed signal design.

Undergraduate Researcher

Feb 2023 – present

University of British Columbia

Vancouver, Canada

- Developed an IEEE 1500 DFT (Design-For-Test) Compliant block for IP developed by the lab.
- Writing a literature review on reliability and wear-out methodologies of IC including electromigration.
- Researched chips that contain more than one reliability sensor to compare with our novel **Sensor-on-Chip**.
- Collaborated with Prof. Andre Ivanov and Ian Hill to author the literature review for publication.

Firmware & Simulation Intern and Researcher

May 2023 – August 2023

TRIUMF

Vancouver, Canada

- Utilized C++ programming language on a **Linux** system to design and optimize **simulation** code for efficient and accurate computation of particle trajectories, energy deposition, and bubble formation.
- Collaborated with a team of researchers at TRIUMF, Snolab, and Fermilab to develop and implement simulation models for dark matter experiments using C++ and Geant4. This includes authoring a research paper together.
- Documented and simulated error sources for the Scintillating Bubble Chamber (SBC).
- Selected as one of three students from a pool of 2000+ applicants for this position.

Integrated Circuits Intern

July 2022 – August 2022

TRYSL Tech

KAUST, Saudi Arabia

- Developed an 8-bit microprocessor using **SystemVerilog** and **Synopsys VCS**, achieving 70% functional coverage within two weeks under contract with a company as part of a digital design project.
- Collaborated with a team to design and test different directional antennae using Synopsys tools and VCS to generate wave functions and test inputs/outputs.
- Acquired knowledge in using *Synopsys Sentaurus* to fabricate MOSFETs and improve antenna functionality.

PROJECTS

FPGA Machine Learning Acceleration | *Verilog, Vivado*

July 2024-August 2024

- Implemented a pre-trained neural network in hardware to speed up inference by 300%.
- Wrote an AXI interface module to communicate neural network inputs & outputs between computer and FPGA.

Football Match Computer Vision | *Python, Machine Learning*

July 2024 - August 2024

- Used YOLO CNN and max-pooling algorithms to detect players on a football pitch
- Implemented k-means to identify shirt colors and which team each player is on.
- Calculated possession percentages and distance covered for each player

Robot Summer Competition Winners | *PCB Design, Firmware, C++*

June 2024 - August 2024

- Designed power board PCB with current sense, H-bridges PCB, I/O Board, Sensor boards from scratch for ENPH Robot competition. Placed 1st out of 15 teams across cohort and featured on global news.

PUBLICATIONS

Research: Wafer-Scale Transfer of MXene Films with Enhanced Device Performance via 2D Liquid Intercalation - in Advanced Materials journal.

TECHNICAL SKILLS

Languages: C/C++, Linux, Python, Java, MatLab, LaTeX

Software Tools: NumPy, Pandas, TensorFlow, Linux, GitHub

Electrical: Altium Designer, KiCad, IC Design, FPGA, SystemVerilog, VHDL, Synopsys VCS

AWARDS

Research: Eric Vogt First Year Summer Research Experience Scholar, Saudi Research Science Institute Top Student Award & Top Presenter Award, 2x Beamline4schools top 40

Scholarships: Dean's Honor's List, International Major Entrance Scholarship, Outstanding International Student Scholarship

School: TKS STEM Gold Credentials, University of Vermont Book Scholar Award